

Proposal Outline – Traffic Accident Prediction

Problem Statement

Traffic accidents are a major public safety issue, causing thousands of injuries and fatalities each year. Understanding the factors that contribute to accident severity can help inform policy, improve road safety, and assist emergency response planning. This project aims to predict the severity of traffic accidents based on environmental and situational factors such as weather conditions, time of day, and location.

Dataset

The U.S. Department of Transportation and related open data portals (such as Kaggle's U.S. Accidents dataset [US Accidents \(2016 - 2023\)](#)) provide detailed traffic crash records from across the country. The dataset typically includes:

- Accident severity (target variable)
- Date and time of the crash
- Location (city, state, GPS coordinates)
- Weather conditions (precipitation, visibility, temperature)
- Road conditions (surface type, lighting)
- Contributing factors (driver behavior, environment)

Methods

1.Data Wrangling – Clean and preprocess the dataset by handling missing values, encoding categorical variables, and addressing class imbalance in accident severity.

2.Exploratory Data Analysis (EDA) – Identify trends by location, time, and weather conditions. Use visualization to highlight high-risk scenarios.

3. Modeling – Apply classification algorithms such as logistic regression, random forest, and gradient boosting to predict accident severity.

4.Evaluation – Compare models using metrics such as accuracy, precision, recall, and F1-score to determine the best-performing model.

Expected Outcomes

- ❖ A predictive model capable of estimating the severity of an accident given environmental and situational factors.
- ❖ Insights into key risk factors (e.g., nighttime driving in poor weather).

- ❖ Visualizations (heatmaps, charts) showing high-risk conditions and locations.
- ❖ Recommendations for improving road safety and reducing severe accidents.